Design and Analysis of 3D-Printable Thin-Shell Dome Structures for Extraterrestrial Habitation

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ABSTRACT

Extreme environmental conditions, unusual loadings, and most importantly, the availability of novel construction techniques will likely dictate the form of any extraterrestrial habitat built on Mars. While a habitat could be constructed by astronauts, it is highly preferred for such a structure to already exist when the first humans land on the Martian surface. Thus, automated structure fabrication equipped with 3D-printing technologies that use in situ resources is an intriguing approach to consider. This paper presents an overview of the design and analysis of a dome-shaped Martian habitat that was designed at Northwestern University in collaboration with Skidmore, Owings & Merrill (SOM) as part of NASA's 3D-Printed Habitat Challenge. The structure has a novel composite hemispheric-parabolic dome that is optimized to sustain self-weight and environmental loads, and to be 3D-printed on an inflatable pressure vessel with Marscrete, a Martian concrete manufactured primarily with local Martian regolith and sulfur. This study examines the structural performance of such a habitat under expected Martian loading conditions, including wind, regolith deposition from storms, and gravity. Furthermore, the habitat performance is assessed under meteorite impact of varying masses and velocities. Finally, a construction scheme, potential internal layout, and functional usability of spaces are also envisioned for a four people unit.

INTRODUCTION

Future missions to Mars will require substantial habitats for their extraterrestrial explorers. Unlike the moon landings, and associated Lunar Lander, Martian missions will carry a prolonged timeline and thus require significant semi-permanent structures. At 34 million miles, material transit for such a structure becomes cumbersome and increasingly difficult (NASA, 2019a). Consequently, in addition to developing a structure to withstand the unique Martian environment for significant duration, it is also desirable to design such a structure within the lens of utilization of locally-sourced, *in situ*, materials and with automated fabrication techniques.

Launched in 2015, NASA's 3D-Printed Habitat Challenge was created to address such concerns, specifically: habitat lifespan, utilization of *in situ* resources, and constructability via 3D-printing. Further specified, the habitat design submissions were required to sustain four astronauts for a period of one Earth-year, provide for 93 square meters of living space, and include all necessary mission spaces (kitchen, sanitation, bedrooms, workspaces, laboratory, communications, and exercise area). Constraints were additionally placed on hatches and exterior penetrations, and designs were required to include mechanical, electrical, plumbing

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(MEP) and environmental control and life support system (ECLSS) equipment (NASA and Bradley University, 2019).

DESIGN BASIS

The greatest drivers for the overall habitat form were constructability, usability, and functionality. For constructability via 3D-printing it is desired to minimize overhanging regions (which would require the use of supports) and maximize continuity (for efficient printing and increased structural integrity by reducing `cold joints'). For usability, this means creating a form which is protective in nature to its inhabitants and suitable for maintaining an internal habitable environment. Finally, a functional habitat is one which satisfies the physical needs of Martian astronauts and provides a level of comfort and livability such to maintain sound mental state for a period of one year.

Several habitat forms meet each of these requirements; however, an internal hemispherical dome with outer parabolic dome structure was found to best satisfy all needs. Multiple self-contained geometries were initially considered (such as pyramidal, conical, cylindrical, and prismatic structures); however, in any such case a near-horizontal surface is required - which demands supports via printing - or an angle of inclination such that an excessive amount of printing material is required. Given that supports become a seeming necessity, this need was embraced, and explored for options as a benefit as opposed to a negative requirement.

Thus, the internal features of the structural shell were designed to exactly match the natural form of an underlying hemispherical pressure vessel, so as to use the pressure vessel for both atmospheric control and construction support. As Northwestern's Marscrete (Wan et al., 2016; Troemner and Cusatis, 2019) was selected for the building material, a design embracing quasi-brittle material behavior is desired. So, the exterior form was fixed as a parabolic dome, such that most external loading conditions result as internal compressive - rather than tensile - stresses. Maintaining these two general forms, several mass optimization iterations were performed on thicknesses ranging from 20 to 150 cm, and this regular hemispherical dome was refined to a thin shell, with varying thickness (21 to 108 cm), and internal cross beams. The final exterior form is shown in Figure 1.

CONSTRUCTION METHOD

To construct the habitat, autonomous rovers are envisioned to clear the site of debris and prepare a shallow cut for the foundation. After site preparation, a foundation is initially 3D-printed utilizing an articulated arm-type 3D-printer. Following foundation printing, a collapsible dome pressure vessel is deployed to the foundation and inflated.

This pressure vessel now doubles as a backing for printing and removes the need for additional printed supports. Reaching to print over such a large inflatable is not desirable as this greatly increases the required size of printer. So, a novel printing scheme was created (as shown in Figure 2) such that the articulating arm need only reach the apex of the habitat, where a small crawler robot is deployed to then circumnavigate the pressure vessel for actual printing. Proceeding from bottom to top is a continuous process (aside from skipping external penetrations), until it reaches the peak, where the arm deploys and affixes an oculus. After printing, the pressure vessel is slightly deflated to provide an offset from the structural Marscrete; allowing for independent thermal deformations of the two bodies.

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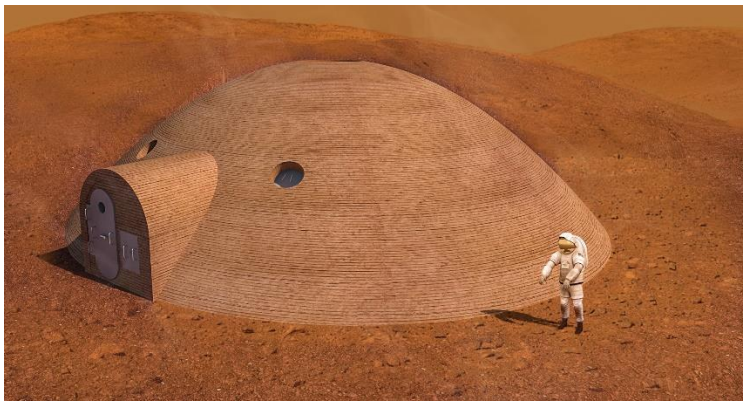

Figure 1: Exterior view of proposed habitat with NASA Z-2 spacesuit for scale.

Figure 2: Demonstration of proposed habitat construction technique, including 'crawler' 3D-printing apparatus.

ENVIRONMENT AND LOADING

While void of Earthly conditions, the Martian environment actually boasts significantly positive characteristics for human habitat construction. At only 38% the gravity of Earth, 3.71 m/s^2 at the equator, design self-weight of structures on Mars is greatly reduced in comparison to Earth construction (Hirt et al., 2012). Further, despite winds from frequent dust storms, given the low air pressure the associated wind load is nearly null. According to NASA, the strongest Martian storms produce winds under 60 mph (NASA, 2015). So, utilizing a conservative estimate of 70 mph and atmospheric density of 0.020 kg/m^3, the expected wind load is calculated as 9.79 Pa. In case of increased density due to airborne dust particles, this load was increased to 60.0 Pa for analysis. The most significant load for design therefore is possible unbalanced sand buildup/dunes from dust storms. With a regolith density of 1,500 kg/m^3 (Peters et al., 2008) and sand depth taken as 0.5 m, a static pressure load of 2,780 Pa was applied to one half of the external structure (assuming unidirectional, unbalanced dust buildup). An additional concern on Mars is a larger-than-typical (for Earth) risk of meteorite impact. While impractical to design for all possible meteorite impacts, it is important to quantify and assess accepted risk. It is also necessary to note an inherent radiation load, and the associated effect on inhabitants; however,

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that concern will not be addressed in this paper as it does not directly influence structural performance.

LOAD ANALYSIS

A finite element static analysis was performed on the habitat using a commercial finite element solver, for the load cases of self-weight, wind pressure, and unbalanced sand buildup. Material properties were assigned such to match published Marscrete behavior (Wan et al., 2016), and the structural response was examined as both a shell- and solid-element model. In both cases, and under all loads (individually and in typical ASCE-7 load combinations), the structural response remained in the elastic region of the material, and with deformation less than 1.0 mm. Additional information on Marscrete strength and behavior can be found in Wan et al. (2016), Troemner and Cusatis (2019), and Troemner et al. (2020).

METEORITE IMPACT PERFORMANCE

Unlike expected loading conditions, it is increasingly difficult to design a Martian habitat to sustain rare meteorite impact loads. While some small meteorites could be accounted for and designed around, as meteorite mass increases, the material and structure required for survival becomes exceedingly impractical. Thus, the proposed habitat design basis was on the aforementioned conditions, and only assessed post-design for meteorite performance under a statistical risk of failure.

Utilizing expected meteorite impact velocity and mass from Mars atmospheric loss and deceleration data (Dycus, 1969), a worst-case-scenario impact is examined, progressively stepping up meteorite masses/velocities until habitat failure. From a Lattice Discrete Particle Model (LDPM) simulation of the proposed habitat shell (Cusatis et al., 2011a,b), it was determined that a surface-normal impact of spherical meteorite 7,390 grams in mass (3.5 g/cm^3) and impact velocity 6.29 km/sec is a critical load. If one uses a conservative ablation factor of 0.01 s^2/km^2 and typical atmospheric entry velocity of 11.7 km/s (Wetherill, 1989) in Equation 1, one arrives at an atmospheric meteorite entry mass of 12,022 grams (Bland, 2000).

$$m_a = m_e \exp[-3(s_e - s_a)] \tag{1}$$

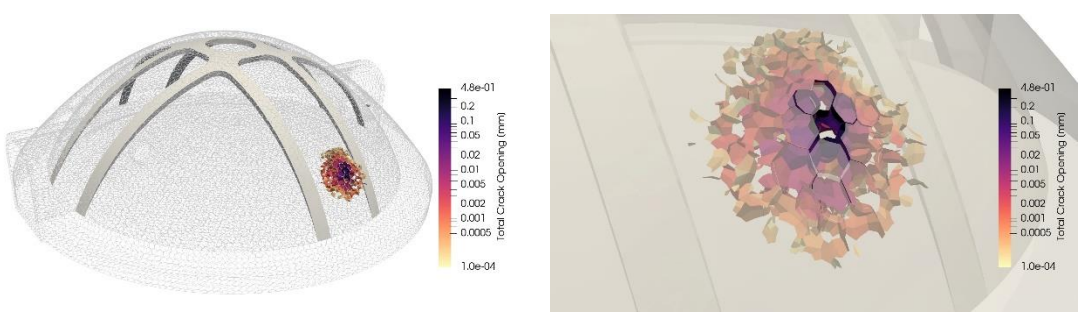

Figure 3: Plot of crack opening resulting from LDPM simulation of impact of largest non-critical meteorite; full habitat shown with cell outlines (left), enlarged impact location (right).

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Given that all meteorites of entry mass 12,022 grams or greater would be considered structurally critical, one is able to determine the atmospheric meteorite flux of hazardous meteorites utilizing Equation 2, also provided in Bland (2000).

$$\log N = -0.689 \log m + 4.17 \tag{2}$$

Here, N is the number of meteorites of mass m or greater entering the atmosphere over a given 10^6 square kilometer area per year. Solving for N arrives at 22.85 meteorites over the area per year. Thus, if the habitat site covers 300 square meters (and conservatively assume all points on-site are equally critical), the probability of a hazardous meteorite impacting the habitat during its one-year mission is approximately 6.9×10^{-9}. As discussed in Bland (2000), entry calculations include a significant level of uncertainty; however, it must be acknowledged that even so, probability of such an impact is exceptionally low.

PROPOSED LAYOUT

With a set exterior form (section view in Figure 4), the interior layout is designed to follow-suit. Inspiration was pulled from the so-called 'tiny house' designs and also the NASA HI-SEAS (Hawaii Space Exploration Analog and Simulation) project. With a greater ceiling height and increased light from the transparent oculus, the center of the habitat is designed as a round common area for recreation and relaxation. Radiating outward from this center area, common and private rooms are arranged according to function and required facilities. The proposed layout, as shown in Figure 5 depicts two semicircular regions - one more public including kitchen, workspaces, and laboratory, and one private including bedrooms and lavatory. Dividing the two semicircular regions is a common passageway connecting two external entries, one with personnel airlock and one with docking equipment for NASA's Multi-Mission Space Exploration Vehicle (MMSEV), or similar vehicle (NASA, 2019b).

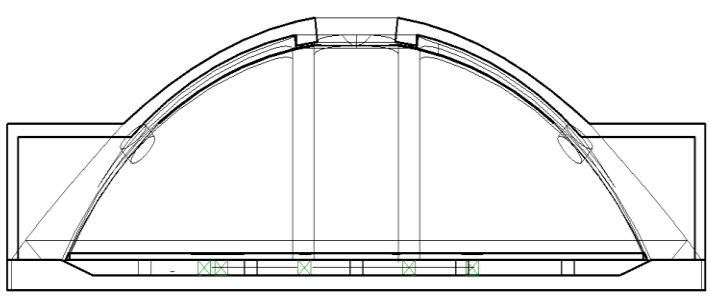

Figure 4: Section view of Martian habitat structure and pressure vessel.

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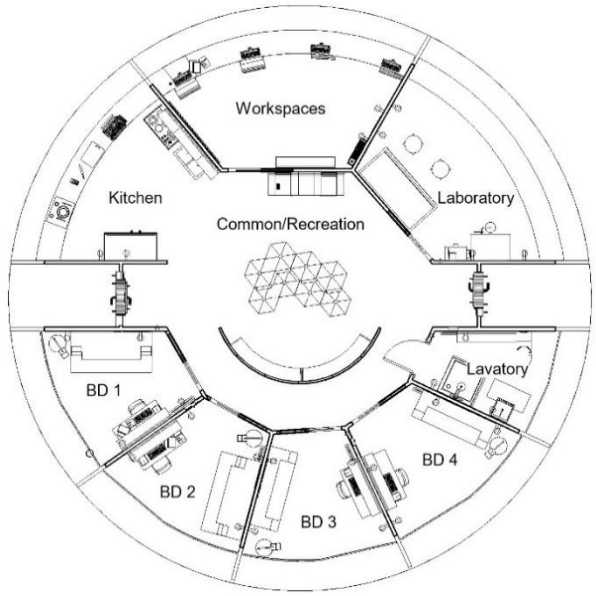

Figure 5: Floor plan of Martian habitat with labels depicting space use.

CONCLUSIONS

In this study a single habitat design was examined for structural performance, constructability, and functionality; and was found to surpass all such hurdles expected of a Martian habitat. The layout offers an environment conducive to astronaut life and achieves all benchmarks set out in NASA's 3D-Printed Habitat Challenge for a one-year mission. The hemispheric-parabolic dome structure employs a unique construction scheme which eliminates the need for printed supports and allows construction via a singular piece of equipment. The structural shell easily performs under all typical Martian loads, while also performing exceptionally well under meteorite impacts up to 7 kg in mass. While designed for the Martian environment, such unique dome structures may pose use-cases on other extraterrestrial sites, the moon, or even Earth.

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